

Fundamentals of Deep Learning and Programming

Lecture 0: Course Overview

Fall 2025

Adjunct Professor: Donghoon Kang

Course Overview

Lecture Time

Fridays, 2 PM – 5 PM

Classroom

Room: L3 301, 302 (3334A, B)

Evaluation Criteria

Attendance (10%)

Programming Assignments (15%)

Term-project (15%)

Mid-term (30%), Final (30%)

- Programming assignments: students should submit the **code** and a **2-3 page document** briefly explaining the code to me by **email** (chocopie@kist.re.kr)

Course Overview

- Absences are **allowed up to two times**, but 2 points will be deducted from the 3rd absence.
- Students who have five or fewer absences and submit all assignments and term projects will receive **at least a B+ grade**, even if their midterm and final exam scores are very low.
- The purpose of this course is not to evaluate students, but to help them **develop basic knowledge and programming skills in deep learning**, a recent trend.

Supplementary Class

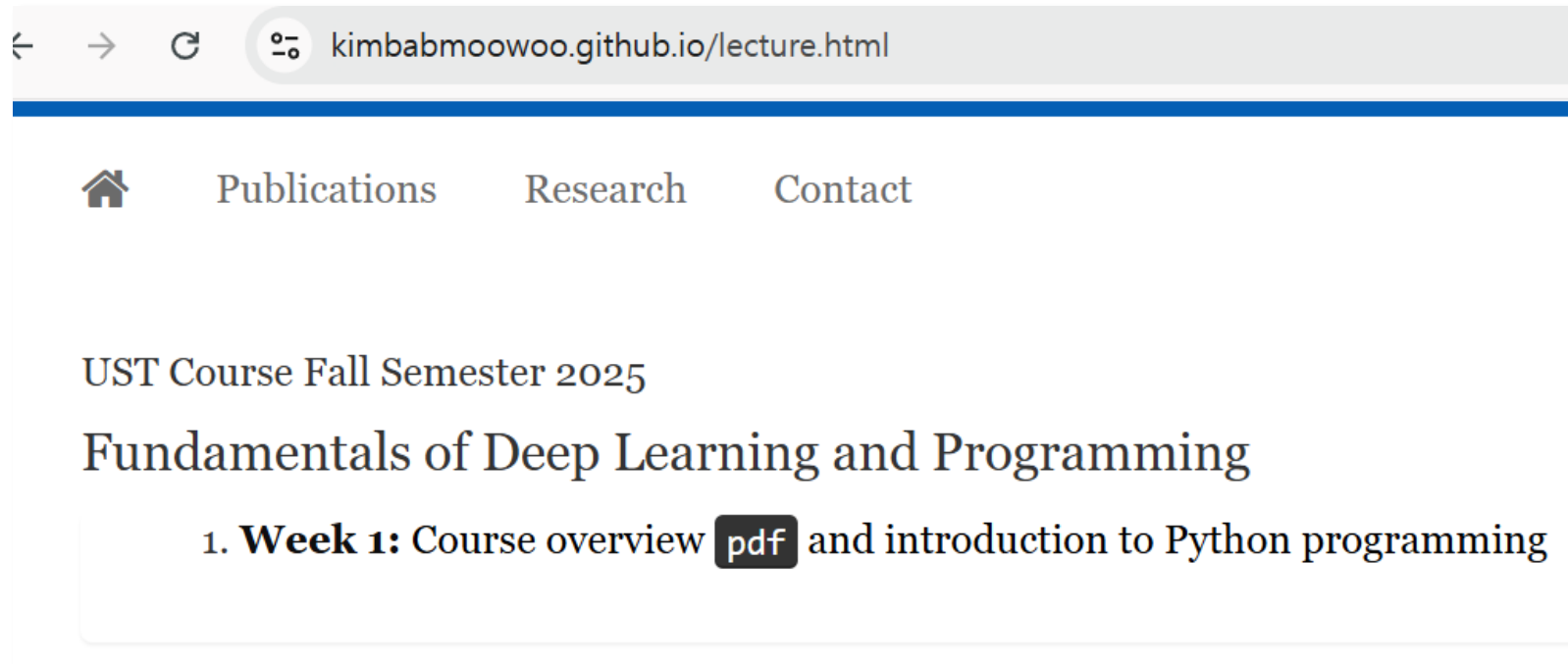
오늘 < 2025.10 > ☐ 음력 ☐ 손없는날 ☒ 기념일						
일	월	화	수	목	금	토
28	29	30	1 국군의 날	2 노인의 날	3 개천절	4 이산가족...
5 세계 한인...	6 추석 음 8.15	7	8 대체 휴일 한로 재향 군인...	9 한글날	10 임산부의 날 정신건강...	11 호스피스...
12	13	14	15 체육의 날	16 세계 식량... 부마민주...	17	18 문화의 날
19	20	21 경찰의 날 음 9.1	22	23 상강	24 국제연합일	25 독도의날
26	27	28 금융의 날 교정의 날	29 지방자치...	30	31 회계의 날	1

Holiday 10.03 (Friday)

Thursday 4 PM ~ 6:30 PM (?)

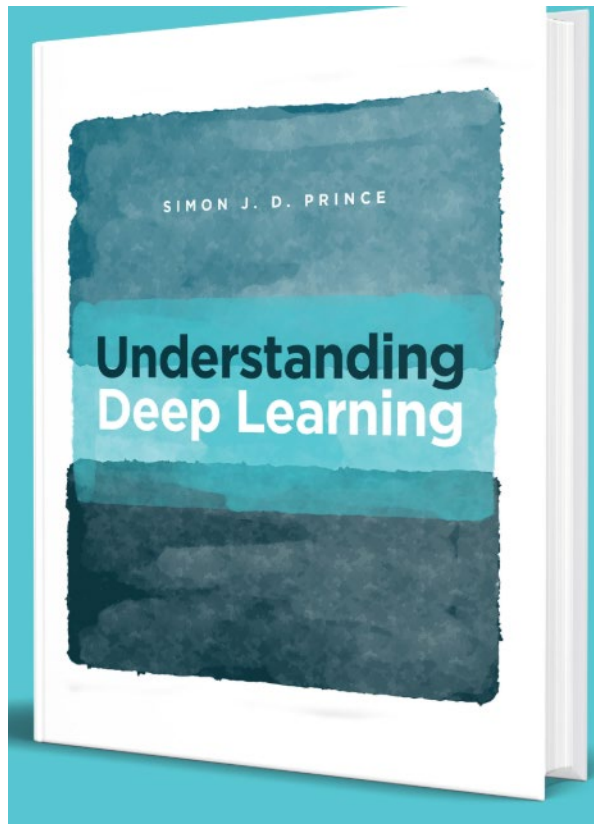
Course Website

<https://kimbabmoowoo.github.io/lecture.html>



Recommended book (pdf is free)

<https://udlbook.github.io/udlbook/>



Understanding Deep Learning

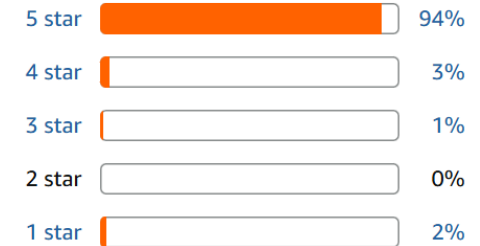
by Simon J.D. Prince | Dec 5, 2023

★★★★★ 168

Customer reviews

★★★★★ 4.9 out of 5

168 global ratings



[Download full PDF \(29 May 2025\)](#)

downloads 523k

Term Project

Must be performed **alone**

- Should **select a paper** published only in **CVPR, ICCV, ECCV, or NeurIPS after 2022 (inclusive)**.
- Should **write an 8-10 page report analyzing the "training code"** of the selected paper.

Before Week 7

Should decide which **paper** to **choose**. (also possible to change another paper later)

Before Week 15

Should **submit** an 8-10 page **report** as a term-project.

Schedule

September

Week 1

Course overview

Introduction to Python Programming

Week 2

Useful Python Libraries

Numpy, PyTorch

Week 3

Neural Networks Basics

Week 4

Convolutional Neural Networks (CNNs)

Schedule

October

Week 5

Models using CNN

Week 6

Introduction to Natural Language Processing (NLP)

Week 7

Attention Mechanism

Week 8

Transformer

Part 1

Week 9

Mid term exam

Schedule

November

Week 10

Transformer

Part 2

Week 11

Transformer

Part 3

Vision Transformer (ViT)

Week 12

Generative Models – Introduction to GANs, VAEs

Week 13

Diffusion Models

Part 1

Schedule

December

Week 14

Diffusion Models

Part 2

Week 15

Diffusion Models

Part 3

Week 16

Final exam